Design Detail Report
Team Phoenix

Kung Fu Quest

College: *IIT PATNA*
Team Leader: *Anand Kumar Pandit*
E-mail: *anand_2401cb42@iitp.ac.in* | *Mobile:* *+91 7050342098*

I. INTRODUCTION

The ABU Robocon 2026 problem statement challenges teams to compete in a strategic game of Tic-Tac-Toe, where two robots operate collaboratively to assemble weapons and place Kung-Fu Scrolls (KFS). One robot (R1) operates in a semi-automatic mode, while the second robot (R2) is fully autonomous. The competition demands rapid weapon assembly, robust autonomous navigation through the Meihua Forest, and precise KFS placement in the Arena. Our design philosophy prioritizes speed, mechanical reliability, and efficient autonomous navigation to achieve a decisive advantage.

II. STRATEGY AND GAME FLOW

After extensive analysis of the ABU Robocon 2026 rulebook, our strategy was formulated to maximize scoring within the 3-minute match duration.

1) **Weapon Assembly (Martial Club):** R1 retrieves and stabilizes the Staff, while R2 retrieves the Spearhead. High-speed actuators ensure weapon assembly is completed within 5 seconds.
2) **Autonomous Navigation (Meihua Forest):** R2 autonomously traverses blocks numbered 1–12 using high-precision encoders combined with LiDAR and vision.
3) **Endgame (Arena – Tic-Tac-Toe):** R1 and R2 coordinate to place KFS strategically to achieve the *"Kung Fu Master"* victory condition.

III. DESCRIPTION OF ROBOT 1 (R1)

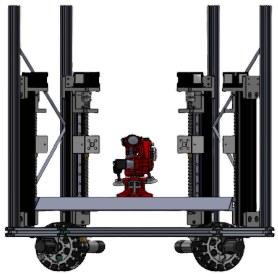
A. *Functional Overview :*

Robot 1 (R1) acts as the logistics anchor. Its core objective is to act as a semi-automatic lift hub for Robot 2 (R2) and a high-torque transport platform for weapon assembly. Unlike standard designs, R1 is engineered to transition from a high-speed transit vehicle on flat zones to a static, structural base capable of hoisting R2 to a height of 1 meter. This functionality is critical for accessing the top row of the Tic-Tac-Toe rack.

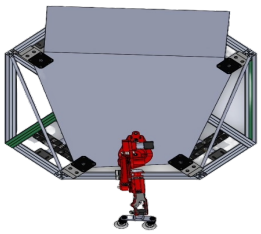
B. *Design Philosophy :*

The architectural design ensures that R1 acts as an immovable object during lifting and an unstoppable force during transit. To resolve the physical conflict between high-speed mobility and the stability required to lift an 18 kg payload, the design employs active Center of Mass (CoM) management. Rather than relying on a static frame, R1 utilizes a dynamic counterweighting system where battery banks shift forward on linear slides to counterbalance the rear-mounted elevator load. Furthermore, deployable pneumatic outriggers widen the support polygon during the stationary lift phase, ensuring zero tipping risk while adhering to the compact starting volume.

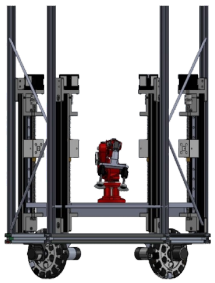
C. *CAD Views :*



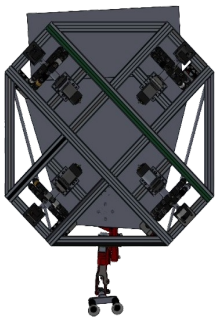
(a) Back View



(b) Top View

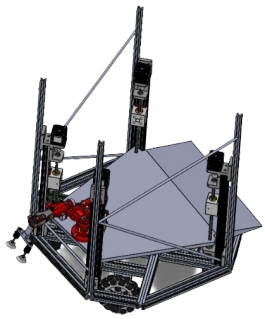


(c) Front View



(d) Bottom View

CAD Views of Robot 1 (Part 1)



(a) Isometric View

(b) Complete Assembly

CAD Views of Robot 1 (Part 2)

IV. DESCRIPTION OF ROBOT 2 (R2) :

A. *Functional Overview :*

Robot 2 (R2) operates as a fully autonomous rover designed under a doctrine of *"Traversability First."* Its mission profile is split between precision weapon assembly in the Martial Club and independent navigation in the Meihua Forest. Unlike standard wheeled platforms that fail on 200 mm obstacles, R2 utilizes a specialized skid-steer track system to traverse irregular terrain. The robot is equipped with a lightweight carbon-fiber telescoping arm capable of a 650 mm extension, allowing accurate placement of KFS blocks on the middle and top tiers of the game rack once elevated by R1.

B. *Design Philosophy :*

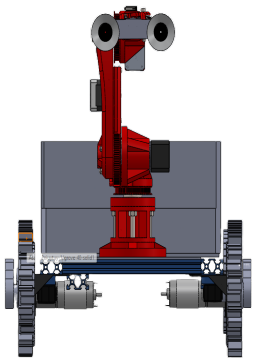
The architectural focus of R2 is defined by geometric stability and torque optimization. To eliminate the risk of back-flipping while climbing 200 mm steps, the chassis length is extended to 720 mm, mechanically enforcing a low Center of Mass (CoM) height of 175 mm. A three-wheel Rocker-Bogie suspension with a 3 mm middle-wheel drop (Δh) is employed. This configuration reduces the effective wheelbase to approximately 360 mm on hard surfaces such as the Arena floor, cutting turning torque requirements by over 50% while ensuring full track engagement on soft terrain. To satisfy the 800 mm starting limit, the intake mechanism utilizes a retractable four-bar linkage that deploys automatically at match start.

The maximum clearable height H_{max} is determined by the sprocket radius r and track inclination θ:

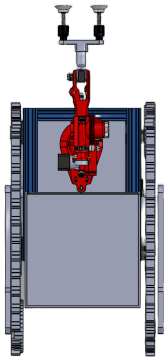
$$H_{max} = r + \left(\frac{L}{2}\right)\sin\theta - \frac{r}{\cos\theta} \tag{IV.1}$$

For our selected radius $r = 70$ mm and optimal approach angle $\theta \approx 64°$, we achieve $H_{max} \approx 233.9$ mm, providing a safety margin of 33.9 mm over the 200 mm obstacle.

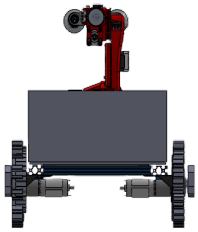
C. *CAD Views :*



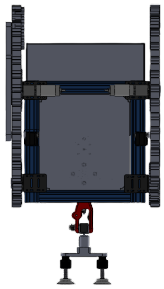
(a) Front View



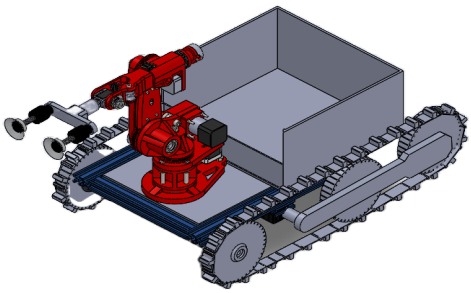
(b) Top View



(a) Back View



(b) Bottom View



(c) Isometric View

CAD Views of Robot 1

V. KFS PICKING & PLACING MECHANISM

A. *Mechanism Description :*

To manipulate the porous Kung Fu Scrolls (KFS) and rigid Staffs, the robots employ a Hybrid Multi-Point Vacuum Gripper mounted on a 6-DOF arm. Instead of a single large interface, this design utilizes a cluster of Four Bellows Suction Tubes (Cups) arranged in a square pattern.

- **Vacuum Source:** A high-flow ejector driven by the 600 kPa air supply converts positive pressure into high-volume suction flow.
- **Interface:**

1) For KFS (Box): All four cups engage the flat surface. The bellows design compensates for surface irregularities.
2) For Weapons (Staffs): The cups are spaced to straddle the 20mm Staff. We utilize Oval Vacuum Cups specifically designed to seal against cylindrical curvatures, preventing the leakage common with standard round cups on narrow pipes.

[**Animation Views :** R1 KFS Picking , R2 KFS Placing]

B. *Calculations and Justifications :*

1. Why 40 kPa Vacuum Limit? Although the rulebook allows 600 kPa positive supply, vacuum is limited to atmospheric pressure (101 kPa). For porous cardboard, leakage through the material prevents deep vacuum. Empirical data suggests a sustainable dynamic limit of $\Delta P \approx 40$ kPa (0.04 N/mm^2).

2. Sizing the Suction Tubes To hold the 0.756 kg KFS at 20 m/s^2 acceleration with a Safety Factor of 4.0, we require a holding force of 90.1 N.

$$A_{total} = \frac{F_{req}}{\Delta P} = \frac{90.1}{0.04} \approx 2252 \text{ mm}^2 \tag{V.1}$$

To achieve this total area using discrete tubes:

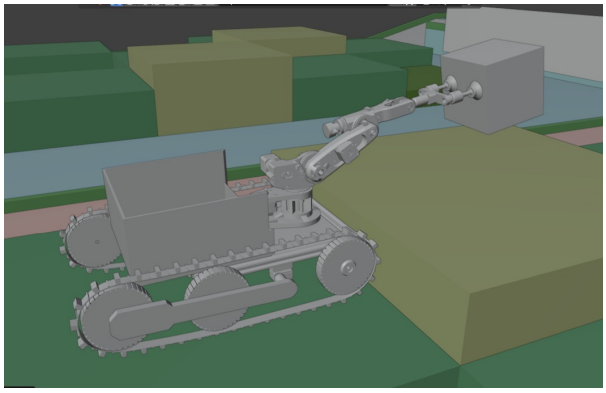
- Proposed Solution: 4 × 30 mm Bellows Cups
- Area per Cup: $A = \pi \cdot (15)^2 \approx 706$ mm^2
- Total Area: $4 \times 706 = 2824$ mm^2

Validation:

$$2824 \text{ mm}^2 > 2252 \text{ mm}^2 \tag{V.2}$$

This configuration provides sufficient holding force while reducing the end-effector footprint compared to a large foam block, allowing for precise placement in the crowded rack.

C. *CAD Model Views :*



CAD representation of the pick-and-place mechanism during KFS handling in the arena.

VI. LIFTING MECHANISM

A. *Mechanism Description and Animation Views :*

Robot 1 (R1) is equipped with an integrated vertical lifting platform designed to elevate Robot 2 (R2) during the endgame phase for high-level Kung Fu Scroll placement. The lifting mechanism consists of a rigid base plate mounted on four linear actuators positioned symmetrically at the four corners of the chassis. These actuators provide synchronized vertical motion, ensuring smooth and stable lifting without tilting.

In the initial configuration, the base plate remains at its lowest position to satisfy the compact starting volume constraints. A small guiding ramp is attached to the platform, allowing Robot 2 to climb onto Robot 1 easily when the plate is lowered. Once Robot 2 is fully positioned on the base plate, the actuators extend simultaneously, lifting the platform vertically and raising Robot 2 to the required height.

This mechanism transforms Robot 1 into a stable lifting hub, enabling Robot 2 to access the upper tiers of the rack while maintaining structural safety and zero tipping risk. **Animation Views:** Lifting]

B. *Calculations and Justifications :*

The lifting mechanism is required to safely elevate Robot 2, whose approximate mass is 18 kg. The total gravitational load acting on the lifting platform is:

$$W = mg = 18 \times 9.81 \approx 176.6 \ N \tag{VI.1}$$

Since the platform is supported by four actuators, the load is equally distributed:

$$F_{\text{actuator}} = \frac{W}{4} = \frac{176.6}{4} \approx 44.15 \ N \tag{VI.2}$$

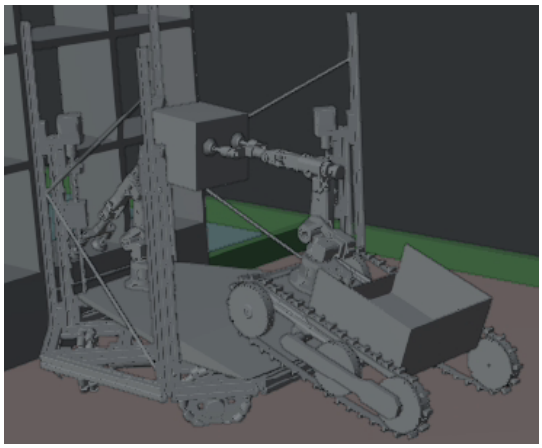
To ensure reliable operation under dynamic match conditions, a safety factor of 2 is applied:

$$F_{\text{design}} = 2 \times 44.15 \approx 88.3 \ N \tag{VI.3}$$

Thus, each actuator must be capable of providing at least 90 N of lifting force. The four-actuator configuration ensures balanced elevation, reduces bending moments on the base plate, and prevents uneven lifting.

C. *CAD Model Views During Lifting :*

The CAD model illustrates the lifting platform mounted on four vertical actuators with the guiding ramp provided for Robot 2 entry. During operation, Robot 2 climbs onto the lowered base plate, after which the actuators extend upward to elevate the platform smoothly.



CAD Model View of R1 Lifting Platform with Ramp for R2 Climbing

VII. OBJECT DETECTION & NAVIGATION

A. *System Description :*

1) Object Detection System: R2 employs a two-stage vision pipeline powered by an NVIDIA Jetson Nano. First, a YOLOv8-Tiny model is utilized to localize and extract the symbol icon from the complex background of the KFS block. Following extraction, a custom-tuned MobileNetV3-Small CNN classifier determines whether the symbol is a valid Oracle Bone script or a Fake pattern. This architecture is exported to ONNX format (2.5 MB) and accepts 1-channel grayscale input to prioritize stroke geometry over color, ensuring robustness across Red and Blue KFS variants.

2) Algorithm: The detection logic follows a sequential process optimized for low latency (≈ 4.1 ms):

1) Icon Extraction (YOLO): The system identifies the KFS block and crops the specific region of interest containing the symbol, effectively acting as a 1.6× digital zoom to prevent resolution loss at distance.

2) Classification (CNN): The cropped image is fed into the MobileNetV3 model. We employ deterministic geometric augmentation (0°, 90°, 180°, 270°) to handle the fixed orientations of the cubic blocks.
3) Verification: The classifier uses an "Active Learning" threshold derived from human-in-the-loop training to reject "Hard Negatives" that mimic valid scripts.

B. *Autonomous Navigation :*

1) Navigation Algorithm: Robot 2 (R2) is equipped with an autonomous navigation system based on real-time SLAM. The robot continuously estimates its pose while generating an updated 3D occupancy grid map of the Meihua Forest. For exploration, R2 implements a Frontier-Based Exploration strategy enhanced with Information Gain, where unexplored boundaries are ranked using traversal cost and obstacle density. A hybrid planning framework is adopted for reliable motion: the global planner applies **Dijkstra's algorithm** on the dynamically updated environment graph to compute the minimum-time path toward the nearest detected Kung Fu Scroll, while the local planner (**Dynamic Window Approach**) ensures smooth and collision-free execution. The SLAM objective is expressed as:

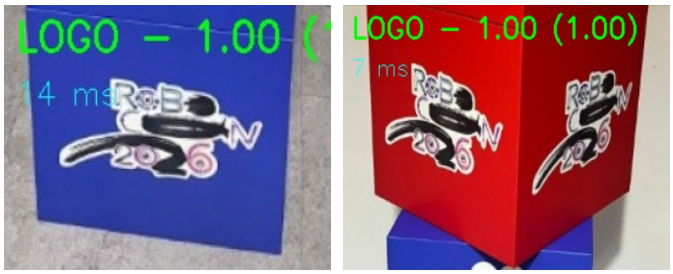
$$x_t, m = \arg\max P(x_t, m \mid z_{1:t}, u_{1:t}) \tag{VII.1}$$

Path planning and velocity control are formulated through global cost minimization and local motion optimization:

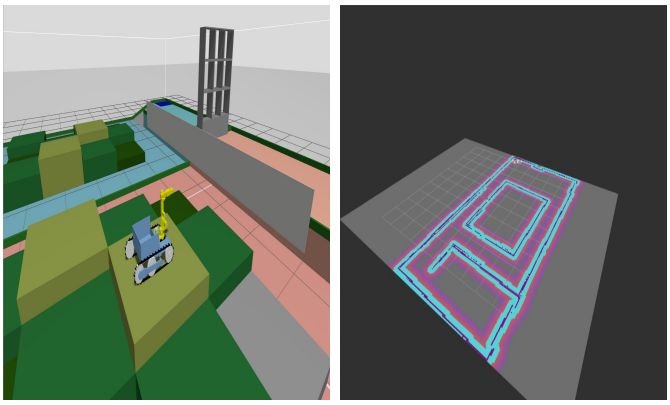
$$(P^*, v^*, \omega^*) = \arg\min_{P} \sum c_{ij} \tag{VII.2}$$

$$(v^*, \omega^*) = \arg\max J(v, \omega) \tag{VII.3}$$

Upon detecting a scroll, R2 retrieves it securely, marks the node as visited, and updates the environmental cost map. This cycle repeats until three scrolls are collected efficiently using the **Nav2** stack and **Google Cartographer** in ROS 2.



Our current KFS Detection Model



(a) Robot in meihua forest (gazebo simulation) **(b)** Cost Map

[Animation Views : Autonomous Navigation **]**

VIII. WEAPON ASSEMBLY MECHANISM

Upon the game start signal, Robot 1 (R1) and Robot 2 (R2) are initiated simultaneously from their respective Start Zones. Both robots operate concurrently throughout the Martial Club phase to achieve rapid weapon assembly.

A. *R1 Staff Picking Mechanism :*

Robot 1 (R1) is equipped with a 5-DoF robotic arm (extendable up to 620 mm) fitted with a detachable, task-specific claw. For staff retrieval, the claw features silicon-based suction grippers that deform according to the target surface. This design provides secure adhesion and significantly reduces positional correction time during pickup.

The flexible gripper also corrects small placement misalignments by conforming to the surface geometry for the correct engagement axis.

B. *R2 Spearhead Picking Mechanism :*

Robot 2 (R2) utilizes a 5-DoF robotic arm with a gripper fitted with two independently controlled vacuum suction cups (radius = 45 mm each). Each cup is powered by its own pneumatic pump, enabling secure grasping of spearheads from the shared Spearhead Rack.

The minimum suction pressure required to lift a spearhead of mass m is derived from the force balance condition with a safety factor S_f:

$$P_{\min} = \frac{mgS_f}{2\pi r^2} \tag{VIII.1}$$

where r is the suction cup radius and the factor of 2 accounts for dual suction cups.

C. *Computer Vision Pipeline for Spearhead Detection :*

To detect and categorize spearheads, R2 employs a vision-based perception pipeline:

- **YOLOv8 Detection:** A pretrained YOLOv8 model detects spearheads and outputs a bounding box ROI, yielding centroid coordinates (x_c, y_c).
- **Image Pre-processing:** The ROI undergoes grayscale conversion, CLAHE-based contrast normalization, and adaptive thresholding for robust segmentation.
- **Shape Classification:** Contour extraction followed by polygon approximation (Ramer-Douglas-Peucker algorithm) identifies the spearhead type (Fist, Palm, or Spear).

D. *Assembly of Staff and Spearhead :*

Once the spearhead centroid (x_c, y_c) and type are determined, R2 aligns the spearhead connector (POM 40SM Quick Coupler) with the staff head held by R1. A Hough Circle Transform is used to locate the circular staff-end centre (x_s, y_s).

Assembly is achieved when the positional alignment satisfies:

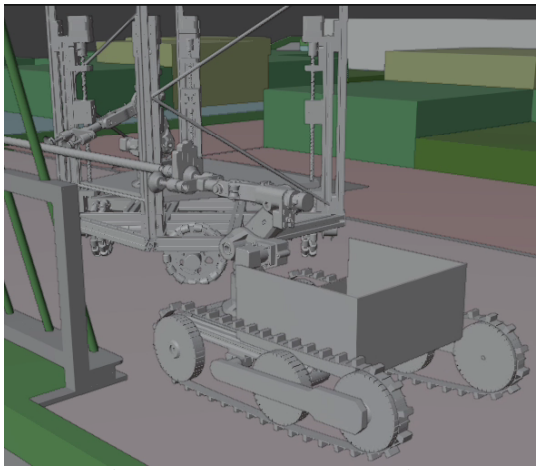
$$\left\| (x_c, y_c) - (x_s, y_s) \right\| \le \varepsilon \tag{VIII.2}$$

where ε is the maximum permissible alignment error.

Each spearhead type has a known canonical orientation. The angular offset $\Delta\theta$ between detected and required orientation is corrected using the wrist joint of the R2 arm:

$$\Delta\theta = \theta_{\text{detected}} - \theta_{\text{canonical}} \tag{VIII.3}$$

R2 rotates the spearhead by $\Delta\theta$ prior to insertion, ensuring correct groove matching and smooth engagement.



Alignment and orientation correction during spearhead-staff assembly.

[Animation Views : Weapon Assembly **]**

IX. ELECTRICAL COMPONENTS

A. *Processing Units :*

1. Nvidia Jetson Orin Nano: Serves as the primary onboard processor for autonomous navigation, SLAM, and real-time environment perception. It processes RGB-D data from the Intel RealSense camera for mapping and obstacle detection, and transmits motion commands to the STM32 via SPI for low-latency control.

2. STM32 Nucleo F446RE: Acts as the dedicated real-time controller running FreeRTOS at 180 MHz. It executes low-level motor tasks such as PID control, PWM signal generation, encoder feedback processing, and actuator synchronization.

B. *Motors and Motor Drivers :*

1. PGM42D775 24V 300 RPM Planetary Gear Motor: We are using this 24V motor with 300 RPM and 20 Kg-cm torque powers movement mechanisms, ensuring weight handling and precise control for navigation for Robot R1.

2. E-BIKE DC Motor MY1016: This motor works at 24V, 2750 RPM motor provides 350W of power for tank mechanism of Robot R2, ideal for lifting bot over obstacles. Lifting custom motors chosen for their lifting capabilities, depending on the bot's load requirements.

3. MD10C Motor Driver: This motor driver supports 5V–30V and up to 13A current (continuous). It provides smooth control for the planetary gear and E-bike motors.

C. *Camera :*

1. Intel RealSense D415: The Intel RealSense D415 is an RGB-D camera that provides simultaneous color and depth data for accurate distance measurement and obstacle detection. It supports 1920×1080 resolution at 30 FPS for RGB only and up to 1280×720 at 30 FPS for depth streaming. It is connected via USB 3.0 to the Jetson Orin Nano which enables reliable SLAM, localization, and path planning for robotic navigation.

2. IMX219-160 8MP IR-Cut Camera: With 8 MP resolution and 120 FPS, this camera provides high-quality vision for identifying real or fake KFS near mechanical claw of Robot R1 and R2.

D. *Proximity Sensors :*

1. Unitree 4D LiDAR L2: It generates high-density 3D point cloud data for SLAM, localization, and obstacle detection. With a wide field of view and very fast refresh rate, it helps to achieve accurate real-time mapping and navigation when integrated with the Jetson Orin Nano using ROS package *cartographer_ros*.

2. Laser Proximity Sensors: It is used for precise short-range distance measurement and alignment. They provide fast, high-accuracy feedback for the precise relative positioning of two bots during weapon assembly and for controlled climbing of R2 onto R1, ensuring accurate docking and stable mechanical engagement.

E. *Communication and Interfaces :*

The Intel RealSense D415 is connected to the Jetson Orin Nano through USB 3.0 and the IMX219-160 camera interfaces through CSI. The Jetson Orin Nano communicates with the STM32 exclusively via SPI for real time control command transmission. The Jetson is connected to an external hand held controller through Bluetooth for controlling robot R1 and Robot R2 is fully autonomous with no Bluetooth or Wi-Fi interface used.

F. *Safety and Compliance :*

All components adhere to the 24V system limit for ABU Robocon 2026 rules. Safety features include overcurrent and thermal protection, emergency STOP buttons, and compliance with competition safety standards.

G. *Integration Workflow :*

- Connect Jetson Orin Nano and STM32 via SPI.
- Interface MD10C motor driver with Jetson Orin Nano via STM32 MCU.
- Set Intel RealSense and IMX219-160 camera on Jetson Orin Nano.
- Assemble and test the custom PCB.
- Test individual components before full integration.

X. CONCLUSION

Our robots for ABU Robocon 2026 have all the necessary mechanisms to ensure maximum chances of success. Our semi-automatic bot (R1) is equipped with:

- Synchronized vertical lifting platform to elevate R2 for high-tier rack access.
- Dynamic counterweight system for active Center of Mass management during heavy lifts.
- 5-DoF arm with a self-aligning claw for rapid Staff retrieval.

Our autonomous bot (R2) uses:

- Skid-steer differential drive mechanism to traverse the obstacles of the Meihua Forest.
- 5-DoF arm with hybrid multi-point vacuum grippers for precise Spearhead and KFS pickup
- Real-time SLAM and a two-stage vision pipeline (YOLOv8 and MobileNetV3) for robust object detection and navigation.

The calculations and results achieved are in accordance with the dimensions and limitations specified in the rule book of ABU Robocon 2026.